

# SPARSE WAVELET ESTIMATION OF ADDITIVE FUNCTIONAL COEFFICIENTS

Author(s)

*Affiliation(s)*

## Supplementary Material

This supplement contains the proofs of the results stated in the paper and the auxiliary lemmas they use. Section S1 fixes the notation and the objects. Section S2 proves the identification results of Section 2 of the paper, Section S3 the approximation lemma, Section S4 the bounds for the Gram matrix of the product design, Section S5 the oracle inequality, and Section S6 the rates, and Section S7 the recovery of the block structure by thresholding. References to numbered results without a section prefix are to the paper; results numbered with the prefix S are stated and proved here.

## S1 Notation and objects

The model, the basis and the estimator are those of Sections 2.1 to 2.3 of the paper. We write  $\mathbf{Z} = [\mathbf{A} \ \mathbf{B}]$  for the design matrix of equation (2.4), with  $\mathbf{A} \in \mathbb{R}^{n \times p}$  the unpenalized block and  $\mathbf{B} \in \mathbb{R}^{n \times d}$ ,  $d = pqN_J$ , the block of products;  $\mathbf{Y}$ ,  $f$  and  $\boldsymbol{\varepsilon}$  for the vectors with entries  $Y_i$ ,  $f(\mathbf{X}_i, \mathbf{U}_i)$  and  $\varepsilon_i$ ;

$\widehat{\Sigma} = \mathbf{Z}^\top \mathbf{Z}/n$  and  $\Sigma = \mathbb{E}(\mathbf{Z}\mathbf{Z}^\top)$ ; and, for  $a, h \in \mathbb{R}^n$ ,  $\|a\|_n^2 = n^{-1} \sum_i a_i^2$  and  $\langle a, h \rangle_n = n^{-1} \sum_i a_i h_i$ .

The [oracle of the approximation space](#) is  $\boldsymbol{\beta}^* = (\mathbf{c}, \boldsymbol{\theta}^*)$ , with  $\theta_{\ell m, jk}^* = \langle g_{\ell m}, \psi_{jk} \rangle$  for  $j < J$ , so that  $f_J = \mathbf{Z}\boldsymbol{\beta}^*$  is the best approximation of  $f$  in the [approximation space](#) in the sense of Lemma 2; the deterministic residual is  $\mathbf{b} = f - f_J$ , the oracle support is  $S_0 = \text{supp}(\boldsymbol{\theta}^*)$  and  $s_0 = |S_0|$ , and  $v = \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*$ . When  $\mathbf{A}$  has rank  $p$  we write  $\Pi_{\mathbf{A}} = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top$ ,  $M_{\mathbf{A}} = I_n - \Pi_{\mathbf{A}}$ ,  $\widetilde{\mathbf{B}} = M_{\mathbf{A}} \mathbf{B}$ ,  $\widehat{\Sigma}_{\mathbf{B}|\mathbf{A}} = \widetilde{\mathbf{B}}^\top \widetilde{\mathbf{B}}/n$  and  $\widetilde{\gamma} = \lambda_{\min}(\widehat{\Sigma}_{\mathbf{B}|\mathbf{A}})$ , and finally  $\widehat{\sigma}_{\max}^2 = \max_a (\widehat{\Sigma})_{aa}$ , the maximum being over the  $d$  penalized columns.

For a support  $S$  and a symmetric matrix  $A$ , the compatibility constant is

$$\phi_0^2(S; A) = \min\{ |S| \delta^\top A \delta : \|\delta_S\|_1 = 1, \|\delta_{S^c}\|_1 \leq 3 \},$$

and the elementary bound  $\delta^\top A \delta \geq \lambda_{\min}(A) \|\delta\|_2^2 \geq \lambda_{\min}(A) \|\delta_S\|_1^2 / |S|$  gives

$$\phi_0^2(S; A) \geq \lambda_{\min}(A) \quad \text{for every } S, \quad (\text{S1.1})$$

so that a lower bound on the smallest eigenvalue is a compatibility bound on every support, with the same constant and no cone restriction.

## S2 Identification and the periodized basis

### Proof of Proposition 1

*Levels and components are determined by the coefficients.* Let  $\beta$  and  $\tilde{\beta}$  be measurable maps from  $[0, 1]^q$  to  $\mathbb{R}^p$  with  $\sum_{\ell} \beta_{\ell}(\mathbf{U}) X_{\ell} = \sum_{\ell} \tilde{\beta}_{\ell}(\mathbf{U}) X_{\ell}$  almost surely, and put  $\delta(u) = \beta(u) - \tilde{\beta}(u)$ . Then  $\{\delta(\mathbf{U})^{\top} \mathbf{X}\}^2 = 0$  almost surely, and since  $\delta(\mathbf{U})$  is measurable with respect to  $\mathbf{U}$  and  $E \|\mathbf{X}\|_2^2 < \infty$ ,

$$0 = E[\{\delta(\mathbf{U})^{\top} \mathbf{X}\}^2 \mid \mathbf{U}] = \delta(\mathbf{U})^{\top} E(\mathbf{X}\mathbf{X}^{\top} \mid \mathbf{U}) \delta(\mathbf{U}) \quad \text{almost surely.}$$

The conditional second moment matrix is positive definite almost surely, and a positive definite quadratic form vanishes only at the origin, so  $\delta(\mathbf{U}) = 0$  almost surely.

*Levels and components are determined within a coefficient.* Because the density of  $\mathbf{U}$  is positive Lebesgue almost everywhere, a set is null for the law of  $\mathbf{U}$  if and only if it is Lebesgue null, so the identity  $c_{\ell} + \sum_m g_{\ell m}(u_m) = \tilde{c}_{\ell} + \sum_m \tilde{g}_{\ell m}(u_m)$  holds Lebesgue almost everywhere on  $[0, 1]^q$ . Write  $\Delta = c_{\ell} - \tilde{c}_{\ell}$  and  $\delta_m = g_{\ell m} - \tilde{g}_{\ell m} \in L_2[0, 1] \subset L_1[0, 1]$ , so that  $\Delta + \sum_m \delta_m(u_m) = 0$  almost everywhere. Each summand is integrable on the cube by Tonelli's theorem. Integrating in all coordinates and using  $\int_0^1 \delta_m = 0$ , which holds because both parametrizations satisfy (2.2), gives  $\Delta = 0$ ; fixing  $m$  and integrating in the remaining coordinates gives  $\delta_m(u_m) = 0$  for almost every  $u_m$ .  $\square$

**Remark S2.1.** If the centering  $\int_0^1 g_{\ell m} = 0$  is replaced by  $E\{g_{\ell m}(U_m)\} = 0$ , the same proof applies under the two sided density bound of Assumption 3, since then  $L_2[0, 1]$  and  $L_2(P_{U_m})$  coincide as sets: the step that uses Fubini's theorem yields  $\delta_m$  constant almost everywhere, and the constant is its mean, which is zero. The two parametrizations are related by  $c_\ell \mapsto c_\ell + \sum_m E\{g_{\ell m}(U_m)\}$ , as stated in Section 2.1 of the paper. The difference is not cosmetic for the estimator: the design of Section 2.3, which omits the constant scaling function, estimates the version centered with respect to Lebesgue measure.

### Proof of Lemma 1

*The basis.* The periodization of the scaling function is  $\phi_{00}^{\text{per}}(u) = \sum_{\iota} \phi(u + \iota)$ , a finite sum at each  $u$  because the support is compact, and it equals the constant function one, since the integer translates of an orthonormal scaling function with  $\int \phi = 1$  form a partition of unity. For a wavelet, exchanging the finite sum with the integral,  $\int_0^1 \psi_{jk} = \int_{\mathbb{R}} 2^{j/2} \psi(2^j y - k) dy = 2^{-j/2} \int_{\mathbb{R}} \psi = 0$  by the vanishing moment of order zero. That  $\{\mathbf{1}\} \cup \{\psi_{jk}\}$  is an orthonormal basis of  $L_2[0, 1]$  is the standard construction of the periodized multiresolution analysis (Daubechies, 1992, Section 9.3). Since  $\dim V_J = 2^J$  and the  $2^J - 1$  wavelets of level below  $J$  are orthogonal to the constant,  $W_J$

is exactly the set of elements of  $V_J$  with zero integral.

*Injectivity.* Suppose two parameter vectors give the same regression function almost surely. The functions  $\beta_\ell^J(u) = c_\ell + \sum_m g_{\ell m}^J(u_m)$ , with  $g_{\ell m}^J = \sum_{jk} \theta_{\ell m, jk} \psi_{jk}$ , are measurable and bounded, so the first part of Proposition 1 gives  $\beta_\ell^J = \tilde{\beta}_\ell^J$  almost surely; each  $g_{\ell m}^J$  has zero integral by the previous paragraph, so the second part gives  $c_\ell = \tilde{c}_\ell$  and  $\sum_{jk} (\theta - \tilde{\theta})_{\ell m, jk} \psi_{jk} = 0$  almost everywhere, whence  $\theta = \tilde{\theta}$  by orthonormality. For the Gram matrix,  $E \|\mathbf{Z}\|_2^2 < \infty$  because  $E \|\mathbf{X}\|_2^2 < \infty$  and each wavelet is bounded, and for  $a \neq 0$ ,  $a^\top \Sigma a = E\{(a^\top \mathbf{Z})^2\} = \|f_a\|_{L_2(P)}^2 > 0$  by injectivity, where  $f_a$  is the regression function with parameter  $a$ .

*Nullity when the scaling column is kept.* With  $\phi_{00} \equiv 1$  the column  $X_\ell \phi_{00}(U_m)$  coincides with the unpenalized column  $X_\ell$ , for each of the  $pq$  pairs, so the  $pq$  vectors that oppose one to the other lie in the kernel and are linearly independent. There are no others: a kernel vector defines  $\sum_\ell X_\ell \{a_\ell + \sum_m (\alpha_{\ell m} + \sum_{jk} a_{\ell m, jk} \psi_{jk}(U_m))\} = 0$  almost surely; by Proposition 1 applied to the part with zero integral,  $a_{\ell m, jk} = 0$  and  $a_\ell + \sum_m \alpha_{\ell m} = 0$ , which is the space of dimension  $pq$  spanned by those vectors.  $\square$

**Remark S2.2** (The interval basis). With the basis of Cohen, Daubechies, and Vial (1993) the wavelets of level  $j \geq j_0$  are orthogonal to  $V_{j_0}$ , which contains the constants, so they still have zero integral and the penalized part

of each block satisfies (2.2) with no restriction. The  $2^{j_0}$  scaling functions do not: the constant is the combination  $1 = \sum_k \mu_k \phi_{j_0 k}^{\text{int}}$  with  $\mu_k = \int_0^1 \phi_{j_0 k}^{\text{int}}$ , and the subspace with zero integral is  $\{\mu^\top \alpha = 0\}$ . Keeping the  $2^{j_0}$  scaling columns of each block unrestricted repeats the constant and makes  $\Sigma$  singular with nullity  $pq$ , as in the last part of Lemma 1; reparametrizing  $\alpha_{\ell m} = Q \gamma_{\ell m}$  with  $Q$  an orthonormal basis of  $\{\mu^\top \alpha = 0\}$  restores injectivity, at the price of  $pq(2^{j_0} - 1)$  unpenalized parameters.

### S3 Approximation by the wavelet spaces $W_J$

#### Proof of Lemma 2

Fix a pair  $(\ell, m)$  and write  $g = g_{\ell m}$  and  $\theta_{jk} = \theta_{\ell m, jk}^*$ .

*Tail sum.* By Step 2 below the coefficients are square summable, so  $g \in L_2[0, 1]$  and, the system being complete and  $g$  orthogonal to the constant,  $g = \sum_{j \geq 0} \sum_k \theta_{jk} \psi_{jk}$  in  $L_2$ . The orthogonal projection onto  $W_J$  retains the levels below  $J$ , and Parseval's identity gives  $\|g - \Pi_J g\|_{L_2[0,1]}^2 = \sum_{j \geq J} \|\theta_{j\cdot}\|_2^2$ .

*From  $\ell_\pi$  to  $\ell_2$  inside a level.* In  $\mathbb{R}^{2^j}$  one has  $\|a\|_2 \leq \|a\|_\pi$  when  $\pi \leq 2$ , and  $\|a\|_2 \leq (2^j)^{1/2-1/\pi} \|a\|_\pi$  by Hölder's inequality when  $\pi > 2$ . In both cases  $\|\theta_{j\cdot}\|_2 \leq 2^{j(1/2-1/\pi)_+} \|\theta_{j\cdot}\|_\pi$ , and Assumption 5 gives

$$\|\theta_{j\cdot}\|_2 \leq C_g 2^{-j\{s+1/2-1/\pi-(1/2-1/\pi)_+\}} = C_g 2^{-js'},$$

since the exponent equals  $s$  when  $\pi \geq 2$  and  $s - (1/\pi - 1/2)$  when  $\pi < 2$ . This is where the loss  $(1/\pi - 1/2)_+$  appears.

*Geometric sum and density.*  $\sum_{j \geq J} C_g^2 2^{-2js'} = C_g^2 2^{-2Js'} / (1 - 2^{-2s'})$ , finite because  $s' > 0$ , which is the first display of Lemma 2. For the second, write  $f - f_J = \sum_{\ell, m} X_\ell h_{\ell m}(U_m)$  with  $h_{\ell m} = g_{\ell m} - \Pi_J g_{\ell m}$ , the levels cancelling. By the Cauchy-Schwarz inequality over the  $pq$  terms and  $|X_\ell| \leq B_X$ ,

$$\mathbb{E}\{(f - f_J)^2\} \leq pq \sum_{\ell, m} \mathbb{E}\{X_\ell^2 h_{\ell m}^2(U_m)\} \leq pq B_X^2 \sum_{\ell, m} \mathbb{E}\{h_{\ell m}^2(U_m)\},$$

and  $\mathbb{E}\{h^2(U_m)\} = \int_0^1 h^2 p_{U_m} \leq C_U \|h\|_{L_2[0,1]}^2$ , the marginal density being bounded by  $C_U$  because integrating the joint density over the other  $q - 1$  coordinates of  $[0, 1]^{q-1}$  does not increase the bound. One factor  $pq$  comes from the Cauchy-Schwarz inequality and the other from the sum.  $\square$

The uniform version is used in Section S5 to bound the empirical norm of the bias without an appeal to Markov's inequality.

**Lemma S3.1** (Uniform approximation error). *Under Assumptions 4 and 5 with  $s > 1/\pi$ , each component has a continuous version and, for every  $J \geq 0$ ,*

$$\|g_{\ell m} - \Pi_J g_{\ell m}\|_\infty \leq \frac{(L - 1) \|\psi\|_\infty C_g}{1 - 2^{-(s-1/\pi)}} 2^{-J(s-1/\pi)},$$

*and therefore  $\|\mathbf{b}\|_n \leq \|f - f_J\|_\infty \leq pq B_X (L - 1) \|\psi\|_\infty C_g 2^{-J(s-1/\pi)} / \{1 - 2^{-(s-1/\pi)}\}$  for every sample.*

*Proof.* At a point  $u$  and a level  $j$  at most  $L - 1$  periodized translates are nonzero, because  $\psi(2^j u - k) \neq 0$  requires  $2^j u - k \in (0, L - 1)$ , so  $\sum_k |\psi_{jk}(u)| \leq (L-1)2^{j/2} \|\psi\|_\infty$ . With  $\max_k |\theta_{jk}| \leq \|\theta_{j\cdot}\|_\pi \leq C_g 2^{-j(s+1/2-1/\pi)}$  the level contributes at most  $(L - 1) \|\psi\|_\infty C_g 2^{-j(s-1/\pi)}$  in the supremum norm. Since  $s > 1/\pi$  the series converges absolutely and uniformly, so its sum is continuous and agrees almost everywhere with  $g$ ; the tail from level  $J$  on is the geometric sum of the displayed bounds, and the statement for  $f - f_J$  is the triangle inequality over the  $pq$  terms.  $\square$

The next proposition is the one referred to in Section 2.2 of the paper: it quantifies what the periodized basis charges for a component that does not match at the two ends of the interval.

**Proposition S3.1** (Cost of periodization). *Let the basis be as in Assumption 4 and let  $g \in B_{\pi,r}^s[0, 1]$ , the Besov space of the interval and not of the circle, with  $s > 1/\pi$ ,  $N > s$  and  $\int_0^1 g = 0$ . Then there is a constant  $C$  depending on  $g$ , on  $\psi$  and on  $(s, \pi)$  such that, for every  $J$  with  $2^J \geq L - 1$ ,*

$$\|g - \Pi_J g\|_{L_2[0,1]}^2 \leq C 2^{-2Js'} + 2(L - 1) \|\psi\|_1^2 \|g\|_\infty^2 2^{-J},$$

*so that the  $L_2$  rate is  $2^{-J \min(s', 1/2)}$ . If  $g$  is continuous with  $g(0) \neq g(1)$ , the second term is of the exact order  $2^{-J}$  and  $\|g - \Pi_J g\|_\infty$  does not converge to zero.*

*Proof.* Let  $\tilde{g}$  be the one periodic extension of  $g$ , so that  $\theta_{jk} = \int_{\mathbb{R}} \tilde{g}(y) 2^{j/2} \psi(2^j y - k) dy$ . At a level with  $2^j \geq L-1$ , separate the interior translates, whose support lies inside  $[0, 1]$ , from the at most  $L-1$  boundary translates that cross the identification of 0 with 1. For an interior translate,  $\theta_{jk} = \langle Eg, \psi_{jk} \rangle_{\mathbb{R}}$  for any extension  $Eg$  of  $g$  to the line; taking  $E$  a bounded extension operator from  $B_{\pi,r}^s[0, 1]$  to  $B_{\pi,r}^s(\mathbb{R})$  (Triebel, 1983, Chapter 3) and using the wavelet characterization of the latter, which for the implication from the function to the sequence requires only  $N > s$  vanishing moments, gives  $\|(\theta_{jk})_{\text{int}}\|_{\pi} \leq C_1 2^{-j(s+1/2-1/\pi)}$ , whose tail sum is bounded as in Lemma 2. For a boundary translate,  $s > 1/\pi$  makes  $g$  bounded and  $|\theta_{jk}| \leq \|g\|_{\infty} \|\psi_{jk}\|_1 = \|g\|_{\infty} \|\psi\|_1 2^{-j/2}$ , and there are at most  $L-1$  of them per level, so their tail sum is at most  $2(L-1) \|\psi\|_1^2 \|g\|_{\infty}^2 2^{-J}$ . For the exact order, if  $g$  is continuous with  $g(0) \neq g(1)$  then on the support of a boundary translate  $\tilde{g}$  equals  $g(1) + o(1)$  to the left of the identification point and  $g(0) + o(1)$  to the right, so  $\theta_{jk} = 2^{-j/2} \{g(0) - g(1)\} \int_{y>a_k} \psi(y) dy + o(2^{-j/2})$ ; since  $\int \psi = 0$  and  $\psi$  vanishes on no subinterval of its support, that partial integral is nonzero for some translate, and the boundary contribution of the level is of order  $2^{-j}$ . The failure of uniform convergence is that of any wavelet series at a jump.  $\square$

Under Assumption 5 nothing changes, since that assumption is already

a periodic one; what Proposition S3.1 gives is the price of its failure. With the basis of Cohen, Daubechies, and Vial (1993) the space  $V_J$  reproduces polynomials of degree below  $N$  and its basis characterizes  $B_{\pi,r}^s[0,1]$  without periodicity (Cohen, 2003, Section 3.9), so Lemma 2 holds with that space and the same proof.

## S4 The Gram matrix of the product design

Recall  $\Psi(u) \in \mathbb{R}^K$ ,  $K = 1 + qN_J$ , the vector that collects the constant and the wavelets of the  $q$  modulators, and the identity  $\mathbf{Z} = \Pi\{\mathbf{X} \otimes \Psi(\mathbf{U})\}$ , where  $\Pi$  is the fixed permutation that moves the  $p$  entries  $X_\ell \cdot 1$  to the front. Since  $\Pi$  is orthogonal, every statement about eigenvalues is invariant to the column order. Assumption 4 also gives a constant  $C_\psi$ , depending only on the family, with

$$\sup_{u \in [0,1]} \sum_{j < J} \sum_k \psi_{jk}(u)^2 \leq C_\psi 2^J \quad \text{for every } J \geq 1 : \quad (\text{S4.1})$$

at each level at most  $L$  periodized wavelets are nonzero at a given  $u$  and  $\|\psi_{jk}\|_\infty = 2^{j/2} \|\psi\|_\infty$ , so the left side is at most  $L \|\psi\|_\infty^2 \sum_{j < J} 2^j \leq L \|\psi\|_\infty^2 2^J$ .

### Proof of Proposition 2

*Part (i), the modulators.* For  $a \in \mathbb{R}^K$  write  $h_a(u) = a^\top \Psi(u) = a_0 + \sum_m h_{a,m}(u_m)$  with  $h_{a,m} = \sum_{jk} a_{m,jk} \psi_{jk}$ . The functions 1 and  $\psi_{jk}(u_m)$ ,  $m = 1, \dots, q$ , are orthonormal in  $L_2([0, 1]^q, du)$ : within one coordinate this is the orthonormality of the periodized system, and across coordinates  $\int \psi_{jk}(u_m) \psi_{j'k'}(u_{m'}) du = \int_0^1 \psi_{jk} \int_0^1 \psi_{j'k'} = 0$  for  $m \neq m'$ , because every wavelet integrates to zero. Hence  $\int_{[0,1]^q} h_a^2 = \|a\|_2^2$  and, by Assumption 3,  $c_U \|a\|_2^2 \leq \mathbb{E}\{h_a(\mathbf{U})^2\} \leq C_U \|a\|_2^2$ , which is the claim since  $a^\top \Sigma_\Psi a = \mathbb{E}\{h_a(\mathbf{U})^2\}$ .

*Part (i), the full matrix.* Let  $\delta \in \mathbb{R}^{pK}$  and reshape it as  $\Delta \in \mathbb{R}^{p \times K}$ , row  $\ell$  holding the coefficients attached to  $X_\ell$ , so that  $\{\mathbf{X} \otimes \Psi(\mathbf{U})\}^\top \delta = \mathbf{X}^\top \Delta \Psi(\mathbf{U})$ . By the tower property,

$$\delta^\top \Pi^\top \Sigma \Pi \delta = \mathbb{E}[\{\mathbf{X}^\top \Delta \Psi(\mathbf{U})\}^2] = \mathbb{E}[\Psi(\mathbf{U})^\top \Delta^\top \mathbb{E}(\mathbf{X} \mathbf{X}^\top \mid \mathbf{U}) \Delta \Psi(\mathbf{U})].$$

Assumption 2 bounds the inner quadratic form between  $\kappa_1 \|\Delta \Psi(\mathbf{U})\|_2^2$  and  $\kappa_2 \|\Delta \Psi(\mathbf{U})\|_2^2$  almost surely, and  $\mathbb{E} \|\Delta \Psi(\mathbf{U})\|_2^2 = \delta^\top (I_p \otimes \Sigma_\Psi) \delta$  lies between  $\lambda_{\min}(\Sigma_\Psi) \|\delta\|_2^2$  and  $\lambda_{\max}(\Sigma_\Psi) \|\delta\|_2^2$ . Combining the two and using the first part gives  $\gamma = \kappa_1 c_U \leq \lambda_{\min}(\Sigma)$  and  $\lambda_{\max}(\Sigma) \leq \kappa_2 C_U$ . If  $\mathbf{X}$  and  $\mathbf{U}$  are independent, the conditional second moment matrix is constant, the same computation gives  $\delta^\top \Pi^\top \Sigma \Pi \delta = \delta^\top \{\mathbb{E}(\mathbf{X} \mathbf{X}^\top) \otimes \Sigma_\Psi\} \delta$  for every  $\delta$ , and two

symmetric matrices with the same quadratic form are equal; the eigenvalues of a Kronecker product of symmetric matrices are the products of the eigenvalues, and  $\lambda_{\min}\{\mathbb{E}(\mathbf{X}\mathbf{X}^\top)\} \geq \kappa_1$  by Jensen's inequality, since  $\lambda_{\min}$  is concave and  $\mathbb{E}(\mathbf{X}\mathbf{X}^\top) = \mathbb{E}\{\mathbb{E}(\mathbf{X}\mathbf{X}^\top \mid \mathbf{U})\}$ .

*Part (ii), the empirical matrix.* First the range:  $\|\mathbf{Z}_i\|_2^2 = \|\mathbf{X}_i\|_2^2 \|\boldsymbol{\Psi}(\mathbf{U}_i)\|_2^2 \leq pB_X^2(1 + qC_\psi 2^J) = R_J$  by (S4.1) applied to each of the  $q$  coordinates. Set  $A_i = (\mathbf{Z}_i\mathbf{Z}_i^\top - \Sigma)/n$ , independent, centered and symmetric, with  $\widehat{\Sigma} - \Sigma = \sum_i A_i$ . Then  $\|\mathbf{Z}_i\mathbf{Z}_i^\top\|_{\text{op}} = \|\mathbf{Z}_i\|_2^2 \leq R_J$  and  $\|\Sigma\|_{\text{op}} \leq R_J$ , so  $\|A_i\|_{\text{op}} \leq 2R_J/n$ ; and  $\mathbb{E}\{(\mathbf{Z}\mathbf{Z}^\top - \Sigma)^2\} = \mathbb{E}(\|\mathbf{Z}\|_2^2 \mathbf{Z}\mathbf{Z}^\top) - \Sigma^2 \preceq R_J\Sigma$ , so that  $\|\sum_i \mathbb{E}A_i^2\|_{\text{op}} \leq R_J\lambda_{\max}(\Sigma)/n \leq R_J\kappa_2 C_U/n$ . The matrix Bernstein inequality (Tropp, 2012, Theorem 1.4) applied to  $\pm \sum_i A_i$  gives, for every  $t > 0$ ,

$$\mathbb{P}(\|\widehat{\Sigma} - \Sigma\|_{\text{op}} \geq t) \leq 2D \exp\left\{-\frac{nt^2}{2R_J(\kappa_2 C_U + 2t/3)}\right\},$$

and  $t = \gamma/2$  with Weyl's inequality gives  $\lambda_{\min}(\widehat{\Sigma}) \geq \lambda_{\min}(\Sigma) - \gamma/2 \geq \gamma/2$  on the complementary event, which is the display of part (ii). The exponent is  $-cn/R_J$  with  $c$  fixed and  $R_J \asymp pq2^J$ , while  $\log(2D) \asymp \log(pq2^J)$ , so the bound tends to zero exactly when  $n/(pq2^J) - \log(pq2^J) \rightarrow \infty$ , which is implied by  $pq2^J \log(pq2^J)/n \rightarrow 0$ . On that event  $\phi_0^2(S; \widehat{\Sigma}) \geq \gamma/2$  for every support, by (S1.1).  $\square$

**Remark S4.1** (What the two assumptions exclude). Assumption 2 fails exactly when a nontrivial linear combination of the linear covariates is al-

most surely a function of the modulators on a set of positive probability; in the canonical case  $X_1 \equiv 1$  and  $X_2 = h(U_1)$  the conditional second moment matrix has rank one and the population Gram matrix degenerates as  $J$  grows, because  $h(u_1)\psi_{jk}(u_1)$  is approached by the span of the basis. In Assumption 3 it is the joint density that is bounded below: if  $U_2 = U_1$  almost surely, both marginals are uniform but  $\psi_{jk}(U_1) - \psi_{jk}(U_2) \equiv 0$  and  $\Sigma_{\Psi}$  is singular. Independence between  $\mathbf{X}$  and  $\mathbf{U}$  buys the exact factorization of part (i) and nothing else.

## S5 The oracle inequality

Four lemmas precede the proof of Theorem 2.

**Lemma S5.1** (Profiling and the Schur complement). *Suppose  $\mathbf{A}$  has rank  $p$ . Then  $\hat{\boldsymbol{\theta}}$  minimizes  $\left\|M_{\mathbf{A}}\mathbf{Y} - \tilde{\mathbf{B}}\boldsymbol{\theta}\right\|_n^2 + 2\lambda\|\boldsymbol{\theta}\|_1$  and  $\hat{\mathbf{c}} = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top(\mathbf{Y} - \mathbf{B}\hat{\boldsymbol{\theta}})$ ; with  $v = \hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*$ ,*

$$\hat{f} - f_J = \tilde{\mathbf{B}}v + \Pi_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon}); \quad (\text{S5.1})$$

$\hat{\Sigma}_{\mathbf{B}|\mathbf{A}}$  is the Schur complement of the block  $\mathbf{A}$  in  $\hat{\Sigma}$  and

$$\tilde{\gamma} = \lambda_{\min}(\hat{\Sigma}_{\mathbf{B}|\mathbf{A}}) \geq \lambda_{\min}(\hat{\Sigma}); \quad (\text{S5.2})$$

and if  $\lambda_{\min}(\hat{\Sigma}) > 0$  the minimizer in (2.5) is unique.

*Proof.* The objective is convex and, for fixed  $\boldsymbol{\theta}$ , quadratic and coercive in  $\mathbf{c}$ , with minimum at the stated value and minimized value  $\|M_{\mathbf{A}}(\mathbf{Y} - \mathbf{B}\boldsymbol{\theta})\|_n^2 = \left\|M_{\mathbf{A}}\mathbf{Y} - \widetilde{\mathbf{B}}\boldsymbol{\theta}\right\|_n^2$ , since  $M_{\mathbf{A}}$  is a symmetric idempotent; minimizing first in  $\mathbf{c}$  and then in  $\boldsymbol{\theta}$  gives the same set of minimizers as minimizing jointly. For (S5.1),  $\mathbf{A}\widehat{\mathbf{c}} = \Pi_{\mathbf{A}}(\mathbf{Y} - \mathbf{B}\widehat{\boldsymbol{\theta}})$  and  $\mathbf{A}\mathbf{c} = \Pi_{\mathbf{A}}\mathbf{A}\mathbf{c}$ , and  $\mathbf{Y} = \mathbf{A}\mathbf{c} + \mathbf{B}\boldsymbol{\theta}^* + \mathbf{b} + \boldsymbol{\varepsilon}$ , so  $\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c}) = \Pi_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon}) - \Pi_{\mathbf{A}}\mathbf{B}v$  and  $\widehat{f} - f_J = M_{\mathbf{A}}\mathbf{B}v + \Pi_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon})$ . For (S5.2), the variational characterization of the Schur complement gives, for every  $v \in \mathbb{R}^d$ ,

$$v^\top \widehat{\Sigma}_{\mathbf{B}|\mathbf{A}} v = \min_{u \in \mathbb{R}^p} (u^\top, v^\top) \widehat{\Sigma} (u^\top, v^\top)^\top \geq \min_u \lambda_{\min}(\widehat{\Sigma}) (\|u\|_2^2 + \|v\|_2^2) = \lambda_{\min}(\widehat{\Sigma}) \|v\|_2^2.$$

Finally  $\lambda_{\min}(\widehat{\Sigma}) > 0$  makes the quadratic term strictly convex, and adding a convex function preserves strict convexity.  $\square$

**Lemma S5.2** (Calibration of the penalty). *Under Assumption 1, for every*

$$\alpha \in (0, 1), \text{ P}\left(\left\|\widetilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n\right\|_\infty > \lambda_0\right) \leq \alpha \text{ with } \lambda_0 = \sigma \widehat{\sigma}_{\max} \{2 \log(2d/\alpha)/n\}^{1/2}.$$

*Proof.* Condition on the covariates, so that  $\widetilde{\mathbf{B}}$  and  $\widehat{\sigma}_{\max}$  are fixed and the errors are independent and sub-Gaussian with parameter  $\sigma$ . For a column  $a$ ,  $(\widetilde{\mathbf{B}}^\top \boldsymbol{\varepsilon})_a/n$  is sub-Gaussian with parameter  $\sigma \|\widetilde{\mathbf{B}}^{(a)}\|_2/n$ , and,  $M_{\mathbf{A}}$  being an orthogonal projection,  $\|\widetilde{\mathbf{B}}^{(a)}\|_2 \leq \|\mathbf{B}^{(a)}\|_2 = \sqrt{n} (\widehat{\Sigma})_{aa}^{1/2} \leq \sqrt{n} \widehat{\sigma}_{\max}$ . With  $t = \lambda_0$  the sub-Gaussian tail bound is at most  $\alpha/d$  per column; a union bound over the  $d$  columns and the tower property finish the proof.  $\square$

**Lemma S5.3** (Empirical column norms). *Under Assumptions 2 to 4, every penalized column has  $E\{(\widehat{\Sigma})_{aa}\} \leq B_X^2 C_U$  and, for every  $t > 0$ , with  $R'_J = B_X^2 \|\psi\|_\infty^2 2^{J-1}$ ,*

$$P(\widehat{\sigma}_{\max}^2 > B_X^2 C_U + t) \leq d \exp\left\{-\frac{nt^2}{2(R'_J B_X^2 C_U + R'_J t/3)}\right\}.$$

*In particular  $P(\widehat{\sigma}_{\max}^2 \leq 2B_X^2 C_U) \rightarrow 1$  whenever  $2^J \log d/n \rightarrow 0$ .*

*Proof.* For the column  $a = (\ell m, jk)$  the summand is  $W_i = X_{i\ell}^2 \psi_{jk}(U_{im})^2 \geq 0$ , with  $E(W) \leq B_X^2 \int_0^1 \psi_{jk}^2 p_{U_m} \leq B_X^2 C_U$  by orthonormality and the bounded marginal density, and  $0 \leq W_i \leq B_X^2 \|\psi\|_\infty^2 2^{J-1} = R'_J$  because  $j \leq J-1$ . Then  $\text{Var}(W) \leq E(W^2) \leq R'_J E(W) \leq R'_J B_X^2 C_U$ , and the one sided Bernstein inequality with a union bound over the  $d$  columns gives the display. With  $t = B_X^2 C_U$  the exponent is  $-3nB_X^2 C_U/(8R'_J)$  and  $R'_J \asymp 2^J$ .  $\square$

**Lemma S5.4** (The  $\ell_1$  norm of the [oracle](#)). *Under Assumption 5, for every pair  $(\ell, m)$  and every level,  $\sum_k |\theta_{\ell m, jk}^*| \leq 2^{j(1-1/\pi)} \|\theta_{\ell m, j}^*\|_\pi \leq C_g 2^{j(1/2-s)}$ , with no dependence on  $\pi$ , and therefore  $\|\theta^*\|_1 \leq pq C_g \sum_{j < J} 2^{j(1/2-s)}$ , which is  $O(1)$  if  $s > 1/2$ , of order  $J$  if  $s = 1/2$  and of order  $2^{J(1/2-s)}$  if  $s < 1/2$ .*

*Proof.* Hölder's inequality in  $\mathbb{R}^{2^j}$  gives  $\|a\|_1 \leq (2^j)^{1-1/\pi} \|a\|_\pi$ , and Assumption 5 bounds the  $\ell_\pi$  norm of the level by  $C_g 2^{-j(s+1/2-1/\pi)}$ ; the resulting exponent is  $j(1-1/\pi) - j(s+1/2-1/\pi) = j(1/2-s)$ , in which  $\pi$  has disappeared. Summing over the levels below  $J$  and over the  $pq$  blocks gives

the three regimes of the geometric sum.  $\square$

### Proof of Theorem 2

*Step 1, the basic inequality.* By Lemma S5.1,  $\widehat{\boldsymbol{\theta}}$  minimizes the residualized objective; comparing its value at  $\widehat{\boldsymbol{\theta}}$  with its value at  $\boldsymbol{\theta}^*$ , and using  $M_{\mathbf{A}}\mathbf{A} = 0$ , so that  $M_{\mathbf{A}}\mathbf{Y} = \widetilde{\mathbf{B}}\boldsymbol{\theta}^* + M_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\epsilon})$ , gives, with  $r = M_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\epsilon})$ ,

$$\left\| \widetilde{\mathbf{B}}v \right\|_n^2 + 2\lambda \left\| \widehat{\boldsymbol{\theta}} \right\|_1 \leq 2\langle r, \widetilde{\mathbf{B}}v \rangle_n + 2\lambda \|\boldsymbol{\theta}^*\|_1. \quad (\text{S5.3})$$

*Step 2, the two noises.* Since  $M_{\mathbf{A}}\widetilde{\mathbf{B}} = \widetilde{\mathbf{B}}$  and  $M_{\mathbf{A}}$  is symmetric,  $\langle r, \widetilde{\mathbf{B}}v \rangle_n = \langle \mathbf{b} + \boldsymbol{\epsilon}, \widetilde{\mathbf{B}}v \rangle_n$ . The stochastic part is handled by duality: on  $\mathcal{T}$ ,  $2|\langle \boldsymbol{\epsilon}, \widetilde{\mathbf{B}}v \rangle_n| \leq 2 \left\| \widetilde{\mathbf{B}}^\top \boldsymbol{\epsilon} / n \right\|_\infty \|v\|_1 \leq \lambda \|v\|_1$ . The deterministic part is handled by the Cauchy-Schwarz and Young inequalities, and not by duality, because  $\mathbf{b}$  is not small coordinatewise:  $2|\langle \mathbf{b}, \widetilde{\mathbf{B}}v \rangle_n| \leq 2\|\mathbf{b}\|_n \left\| \widetilde{\mathbf{B}}v \right\|_n \leq \left\| \widetilde{\mathbf{B}}v \right\|_n^2 / 2 + 2\|\mathbf{b}\|_n^2$ . Substituting both in (S5.3),

$$\frac{1}{2} \left\| \widetilde{\mathbf{B}}v \right\|_n^2 + 2\lambda \left\| \widehat{\boldsymbol{\theta}} \right\|_1 \leq \lambda \|v\|_1 + 2\lambda \|\boldsymbol{\theta}^*\|_1 + 2\|\mathbf{b}\|_n^2. \quad (\text{S5.4})$$

*Step 3, the slow rate.* Bounding  $\|v\|_1 \leq \left\| \widehat{\boldsymbol{\theta}} \right\|_1 + \|\boldsymbol{\theta}^*\|_1$  in (S5.4) gives  $\frac{1}{2} \left\| \widetilde{\mathbf{B}}v \right\|_n^2 + \lambda \left\| \widehat{\boldsymbol{\theta}} \right\|_1 \leq 3\lambda \|\boldsymbol{\theta}^*\|_1 + 2\|\mathbf{b}\|_n^2$ , which is part (i). No property of  $\widehat{\Sigma}$  was used.

*Step 4, the cone.* Since  $\boldsymbol{\theta}^*$  vanishes off  $S_0$ ,  $\left\| \widehat{\boldsymbol{\theta}} \right\|_1 \geq \|\boldsymbol{\theta}^*\|_1 - \|v_{S_0}\|_1 + \|v_{S_0^c}\|_1$  and  $\|v\|_1 = \|v_{S_0}\|_1 + \|v_{S_0^c}\|_1$ ; substituting in (S5.4) and cancelling

$$2\lambda \|\boldsymbol{\theta}^*\|_1,$$

$$\frac{1}{2} \left\| \tilde{\mathbf{B}}v \right\|_n^2 + \lambda \|v_{S_0^c}\|_1 \leq 3\lambda \|v_{S_0}\|_1 + 2 \|\mathbf{b}\|_n^2. \quad (\text{S5.5})$$

*Step 5, two cases.* If  $2 \|\mathbf{b}\|_n^2 \leq \lambda \|v_{S_0}\|_1$ , then (S5.5) gives  $\frac{1}{2} \left\| \tilde{\mathbf{B}}v \right\|_n^2 + \lambda \|v_{S_0^c}\|_1 \leq 4\lambda \|v_{S_0}\|_1$ ; by the Cauchy-Schwarz inequality and (S5.2),  $\|v_{S_0}\|_1 \leq \sqrt{s_0} \|v\|_2 \leq \sqrt{s_0} \left\| \tilde{\mathbf{B}}v \right\|_n / \sqrt{\tilde{\gamma}}$ , so  $\left\| \tilde{\mathbf{B}}v \right\|_n \leq 8\lambda \sqrt{s_0/\tilde{\gamma}}$  and  $\|v\|_2^2 \leq \left\| \tilde{\mathbf{B}}v \right\|_n^2 / \tilde{\gamma} \leq 64\lambda^2 s_0 / \tilde{\gamma}^2$ ; the same display gives  $\|v_{S_0^c}\|_1 \leq 4 \|v_{S_0}\|_1$  and hence  $\|v\|_1 \leq 5 \|v_{S_0}\|_1 \leq 40\lambda s_0 / \tilde{\gamma}$ . If instead  $2 \|\mathbf{b}\|_n^2 > \lambda \|v_{S_0}\|_1$ , then (S5.5) gives  $\frac{1}{2} \left\| \tilde{\mathbf{B}}v \right\|_n^2 < 8 \|\mathbf{b}\|_n^2$ , that is  $\left\| \tilde{\mathbf{B}}v \right\|_n^2 < 16 \|\mathbf{b}\|_n^2$ , together with  $\lambda \|v\|_1 < 10 \|\mathbf{b}\|_n^2$  and  $\|v\|_2^2 < 16 \|\mathbf{b}\|_n^2 / \tilde{\gamma}$ . Each quantity is bounded by the maximum of the two cases, hence by their sum, which is part (ii).

*Step 6, prediction.* By (S5.1),  $\hat{f} - f = \tilde{\mathbf{B}}v + \Pi_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon}) - \mathbf{b}$ , and  $\Pi_{\mathbf{A}}$  being a projection,  $\|\Pi_{\mathbf{A}}\mathbf{b}\|_n \leq \|\mathbf{b}\|_n$ , which gives the triangle inequality of part (iii). Conditionally on the covariates the errors are independent, centered and of variance at most  $\sigma^2$ , so

$$\mathbb{E}(\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2 \mid \mathbf{X}, \mathbf{U}) = \frac{1}{n} \text{tr}\{\Pi_{\mathbf{A}} \text{Cov}(\boldsymbol{\varepsilon} \mid \mathbf{X}, \mathbf{U})\} \leq \frac{\sigma^2}{n} \text{tr}(\Pi_{\mathbf{A}}) \leq \frac{\sigma^2 p}{n},$$

since the diagonal entries of  $\Pi_{\mathbf{A}}$  are nonnegative and its trace is the rank of  $\mathbf{A}$ . □

**Remark S5.1** (Why there is no cone condition). The usual treatment of unpenalized coordinates (Bühlmann and van de Geer, 2011, Section 6.9)

places them inside the support and requires compatibility on the cone  $\|\delta_{S^c}\|_1 \leq 3\|\delta_S\|_1$ . Profiling removes them from the problem, and (S5.2) transfers the full smallest eigenvalue of  $\widehat{\Sigma}$  to the residualized block, so that Proposition 2(ii) is used as it stands and the constant does not depend on  $S_0$ . If only the restricted version of the design condition were available, the cone would return, and with the constant 4 rather than 3, since Step 5 produces  $\|v_{S_0^c}\|_1 \leq 4\|v_{S_0}\|_1$ : the bias consumes part of the slack.

**Corollary S5.1** (The two rates with the design condition instantiated).

*Add Assumptions 2 to 5 and suppose  $2^J \log d/n \rightarrow 0$ , which is implied by the requirement of Proposition 2(ii). Take  $\lambda = 2\sigma B_X \sqrt{2C_U} \{2 \log(2d/\alpha)/n\}^{1/2}$ . Then, with probability at least  $1 - \alpha - \alpha' - \alpha''$  up to terms that tend to zero,*

$$\left\| \widehat{f} - f \right\|_n^2 \lesssim \frac{\sigma^2 B_X^2 C_U}{\kappa_1 C_U} \frac{s_0 \log(pqN_J)}{n} + \frac{\mathcal{B}_J^2}{\alpha'} + \frac{\sigma^2 p}{n\alpha''},$$

*and, with no condition on the design, the same bound with  $\lambda \|\boldsymbol{\theta}^*\|_1$  of Lemma S5.4 in place of the first term.*

*Proof.* Intersect four events:  $\mathcal{T}$ , of probability at least  $1 - \alpha$ , by Lemma S5.2;  $\{\widehat{\sigma}_{\max}^2 \leq 2B_X^2 C_U\}$ , by Lemma S5.3, which makes the stated  $\lambda$  admissible, that is  $\lambda \geq 2\lambda_0$ ;  $\{\lambda_{\min}(\widehat{\Sigma}) \geq \gamma/2\}$ , by Proposition 2(ii), which with (S5.2) gives  $\widetilde{\gamma} \geq \gamma/2$ ; and  $\{\|\mathbf{b}\|_n^2 \leq \mathcal{B}_J^2/\alpha'\}$ , which is Markov's inequality applied to  $\mathbb{E} \|\mathbf{b}\|_n^2 = \|f - f_J\|_{L_2(P)}^2 \leq \mathcal{B}_J^2$ , the identity holding because the observations

are identically distributed. Substituting in parts (ii) and (iii) of Theorem 2 gives the statement, and part (i) with Lemma S5.4 gives the unconditional version.  $\square$

## S6 Rates

Throughout this section  $p, q, \sigma, B_X, c_U, C_U, \kappa_1, \kappa_2, C_g, s, \pi, r$  and  $\alpha$  are fixed as  $n$  grows,  $J = J_n \rightarrow \infty$ ,  $d_n = pq(2^{J_n} - 1)$ ,  $L_n = \log(2d_n/\alpha) \asymp J_n$ , and  $\lambda_n$  is as in (3.1). Since the constants of Theorem 2 depend on the confidence levels only through  $1/\alpha'$  and  $1/\alpha''$ , taking  $\alpha' = \alpha'' = \epsilon/3$  gives, for each  $\epsilon > 0$ , a constant  $M_\epsilon$  with  $P(\|\hat{f} - f\|_n^2 > M_\epsilon r_n) < \epsilon$ , which is the meaning of the statements in  $O_p$  below. We also write  $\Lambda = \kappa_2 C_U + \gamma/2$ .

The first lemma is Theorem 2 read at an arbitrary comparator. It consumes no new step: the proof of that theorem uses the oracle only through the decomposition of  $\mathbf{Y}$  and through the size of its support.

**Lemma S6.1** (Theorem 2 with an arbitrary comparator). *Under Assumption 1, with  $\mathbf{A}$  of rank  $p$  and  $\lambda \geq 2\lambda_0$ , on the event  $\mathcal{T}$  the following hold simultaneously for every  $\bar{\boldsymbol{\beta}} = (\bar{\mathbf{c}}, \bar{\boldsymbol{\theta}})$ , with  $\bar{\mathbf{b}} = f - \mathbf{Z}\bar{\boldsymbol{\beta}}$ ,  $\bar{M} = |\text{supp}(\bar{\boldsymbol{\theta}})|$  and  $\bar{v} = \hat{\boldsymbol{\theta}} - \bar{\boldsymbol{\theta}}$ :  $\frac{1}{2} \|\tilde{\mathbf{B}}\bar{v}\|_n^2 + \lambda \|\hat{\boldsymbol{\theta}}\|_1 \leq 3\lambda \|\bar{\boldsymbol{\theta}}\|_1 + 2 \|\bar{\mathbf{b}}\|_n^2$ ; if  $\tilde{\gamma} > 0$ ,  $\|\tilde{\mathbf{B}}\bar{v}\|_n^2 \leq$*

$$64\lambda^2\bar{M}/\tilde{\gamma} + 16 \|\bar{\mathbf{b}}\|_n^2 \text{ and } \|\bar{v}\|_2^2 \leq 64\lambda^2\bar{M}/\tilde{\gamma}^2 + 16 \|\bar{\mathbf{b}}\|_n^2/\tilde{\gamma}; \text{ and}$$

$$\|\hat{f} - f\|_n^2 \leq 3 \min_{\bar{\beta}} \left\{ \frac{64\lambda^2\bar{M}}{\tilde{\gamma}} + 20 \|\bar{\mathbf{b}}\|_n^2 \right\} + 3 \|\Pi_{\mathbf{A}}\epsilon\|_n^2. \quad (\text{S6.1})$$

*Proof.* Follow the proof of Theorem 2 with  $(\bar{\mathbf{c}}, \bar{\boldsymbol{\theta}}, \bar{\mathbf{b}}, \text{supp}(\bar{\boldsymbol{\theta}}), \bar{M}, \bar{v})$  in place of  $(\mathbf{c}, \boldsymbol{\theta}^*, \mathbf{b}, S_0, s_0, v)$ . Step 1 uses only  $\mathbf{Y} = \mathbf{A}\bar{\mathbf{c}} + \mathbf{B}\bar{\boldsymbol{\theta}} + \bar{\mathbf{b}} + \epsilon$ , which is the definition of  $\bar{\mathbf{b}}$ , and  $M_{\mathbf{A}}\mathbf{A} = 0$ . Step 2 uses  $\mathcal{T}$ , which does not depend on the comparator, and  $\|M_{\mathbf{A}}\bar{\mathbf{b}}\|_n \leq \|\bar{\mathbf{b}}\|_n$ . Steps 3 to 5 are algebra with  $\text{supp}(\bar{\boldsymbol{\theta}})$  in place of  $S_0$ , using  $\bar{\boldsymbol{\theta}}_{\text{supp}(\bar{\boldsymbol{\theta}})^c} = 0$ , the Cauchy-Schwarz inequality and (S5.2). Step 6 gives  $\hat{f} - \mathbf{Z}\bar{\beta} = \tilde{\mathbf{B}}\bar{v} + \Pi_{\mathbf{A}}(\bar{\mathbf{b}} + \epsilon)$  and hence  $\hat{f} - f = \tilde{\mathbf{B}}\bar{v} + \Pi_{\mathbf{A}}(\bar{\mathbf{b}} + \epsilon) - \bar{\mathbf{b}}$ . Since  $\mathcal{T}$ ,  $\tilde{\gamma}$  and the condition on  $\lambda$  do not involve  $\bar{\beta}$ , the inequalities hold simultaneously on the same event and the minimum may be taken; (S6.1) follows from  $(x + y + z)^2 \leq 3(x^2 + y^2 + z^2)$ .  $\square$

### Proof of Lemma 3

*The envelope.* Fix  $\epsilon > 0$  and count  $N(\epsilon) = \#\{a : |\theta_a^*| > \epsilon\}$ . Inside a block and a level, Markov's inequality in the  $\ell_\pi$  norm gives  $\#\{k : |\theta_{\ell m, jk}^*| > \epsilon\} \leq \epsilon^{-\pi} \|\theta_{\ell m, j\cdot}^*\|_\pi^\pi$ , and Assumption 5 gives  $\|\theta_{\ell m, j\cdot}^*\|_\pi^\pi \leq C_g^\pi 2^{-j\pi(s+1/2-1/\pi)} = C_g^\pi 2^{j(1-a)}$  with  $a = \pi(s+1/2) > 1$ . Since a level has  $2^j$  coordinates,

$$N_{\ell m}(\epsilon) \leq \sum_{j \geq 0} \min\{2^j, C_g^\pi \epsilon^{-\pi} 2^{j(1-a)}\}.$$

The two arguments agree at  $2^{j_\epsilon a} = (C_g/\epsilon)^\pi$ , that is at  $2^{j_\epsilon} = (C_g/\epsilon)^\tau$ , since  $\pi/a = (s + 1/2)^{-1} = \tau$ . Below that level the minimum is  $2^j$  and the sum is at most  $2^{j_\epsilon}$ ; above it the minimum is the second term, which decreases geometrically with ratio  $2^{1-a} < 1$ , and the sum is  $2^{j_\epsilon}/(1 - 2^{1-a})$ . Adding the two parts and the  $pq$  blocks gives  $N(\epsilon) \leq pqA_{s,\pi}(C_g/\epsilon)^\tau$ ; truncating at  $j < J$  only decreases the count, so the bound holds for every  $J$ . Finally,  $\epsilon < \theta_{(k)}^*$  implies  $N(\epsilon) \geq k$ , hence  $\epsilon \leq C_g(pqA_{s,\pi}/k)^{1/\tau}$ , and letting  $\epsilon$  increase to  $\theta_{(k)}^*$  gives the envelope; and  $\tau = (s + 1/2)^{-1} < 2$  for every  $s > 0$ .

*The tail.* With  $2/\tau > 1$ ,  $\sum_{k>M} \theta_{(k)}^2 \leq C_\tau^2 \sum_{k>M} k^{-2/\tau} \leq C_\tau^2 \int_M^\infty x^{-2/\tau} dx = C_\tau^2 M^{1-2/\tau} \tau / (2 - \tau)$ .  $\square$

We also record that, on the event of Proposition 2(ii) with  $t = \gamma/2$ ,

$$\|\mathbf{B}\delta\|_n^2 \leq \lambda_{\max}(\widehat{\Sigma}) \|\delta\|_2^2 \quad \text{and} \quad \lambda_{\max}(\widehat{\Sigma}) \leq \Lambda, \quad (\text{S6.2})$$

the first because  $\mathbf{B}^\top \mathbf{B}/n$  is a principal submatrix of  $\widehat{\Sigma}$  and the second by Weyl's inequality and Proposition 2(i).

### Proof of Proposition 3

By parts (i) and (iii) of Theorem 2, on  $\mathcal{T}$ ,  $\left\| \widetilde{\mathbf{B}}v \right\|_n^2 \leq 6\lambda \|\boldsymbol{\theta}^*\|_1 + 4\|\mathbf{b}\|_n^2$  and  $\left\| \widehat{f} - f \right\|_n^2 \leq 3(\left\| \widetilde{\mathbf{B}}v \right\|_n^2 + 4\|\mathbf{b}\|_n^2 + \|\Pi_{\mathbf{A}}\boldsymbol{\epsilon}\|_n^2)$ , and no property of  $\widehat{\Sigma}$  enters. Lemma S5.3 makes  $\lambda_n$  admissible with probability tending to one, since  $2^{J_n} \log d_n/n \rightarrow 0$  in every regime considered; Markov's inequality applied

to  $E \|\mathbf{b}\|_n^2 \leq \mathcal{B}_{J_n}^2$ , from Lemma 2, and to  $E(\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2 \mid \mathbf{X}, \mathbf{U}) \leq \sigma^2 p/n$  gives the general statement. If  $s > 1/2$ , Lemma S5.4 gives  $\|\boldsymbol{\theta}^*\|_1 = O(1)$  and  $\lambda_n \|\boldsymbol{\theta}^*\|_1 \asymp (J_n/n)^{1/2}$ , and it suffices that the bias not dominate, that is  $2^{-2J_n s'} \leq (J_n/n)^{1/2}$ , which holds for  $J_n \geq \log_2(n/J_n)/(4s')$ . If  $s < 1/2$ , then  $\|\boldsymbol{\theta}^*\|_1 \asymp 2^{J_n(1/2-s)}$ , the two terms are  $(J_n/n)^{1/2} 2^{J_n(1/2-s)}$  and  $2^{-2J_n s'}$ , and equating them gives  $2^{J_n} \asymp (n/J_n)^{1/(1-2s+4s')}$  with common value  $(J_n/n)^{2s'/(1-2s+4s')}$ ; when  $s' = s$  the denominator is  $1 + 2s$ .  $\square$

### Proof of Theorem 3

By the definition of  $J_n$ ,  $(n/\log n)^{1/(2s'+1)} \leq 2^{J_n} < 2(n/\log n)^{1/(2s'+1)}$  and  $J_n \asymp \log n$ , so

$$\frac{pq2^{J_n} \log(pq2^{J_n})}{n} \asymp \frac{(n/\log n)^{1/(2s'+1)} \log n}{n} = \left(\frac{\log n}{n}\right)^{2s'/(2s'+1)} \longrightarrow 0,$$

which is the requirement of Proposition 2(ii); (S5.2) then gives  $\tilde{\gamma} \geq \gamma/2$  with probability tending to one. On that event Corollary S5.1 with  $s_0 \leq d_n$  gives

$$\left\| \hat{f} - f \right\|_n^2 \lesssim \frac{\sigma^2 B_X^2 C_U}{\kappa_1 c_U} \frac{d_n L_n}{n} + \frac{\mathcal{B}_{J_n}^2}{\alpha'} + \frac{\sigma^2 p}{n \alpha''}.$$

With  $d_n \asymp pq2^{J_n}$  and  $L_n \asymp \log n$  the first term is the display above; the second is  $\mathcal{B}_{J_n}^2 \asymp 2^{-2J_n s'} \leq (\log n/n)^{2s'/(2s'+1)}$  by the lower bound on  $2^{J_n}$ ; and the third is  $O(1/n)$ , of strictly smaller order because  $2s'/(2s'+1) < 1$ .  $\square$

### Proof of Corollary 1

Inside a block the wavelets of level below  $J$  are orthonormal in  $L_2[0, 1]$ , so  $\|\widehat{g}_{\ell m} - \Pi_J g_{\ell m}\|_{L_2[0,1]}^2 = \sum_{jk} v_{\ell m, jk}^2$ , and summing over the  $pq$  blocks gives  $\|v\|_2^2$ ; adding the bias of Lemma 2 in each block,

$$\sum_{\ell, m} \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2 \leq 2\|v\|_2^2 + \frac{2pqC_g^2}{1 - 2^{-2s'}} 2^{-2Js'}.$$

Theorem 2(ii) bounds  $\|v\|_2^2$  by  $64\lambda^2 s_0 / \widetilde{\gamma}^2 + 16 \|\mathbf{b}\|_n^2 / \widetilde{\gamma}$ , and on the event of Theorem 3 the quantity  $\widetilde{\gamma}$  is bounded below by the constant  $\gamma/2$ , so the two terms are of the same order as those of Theorem 3, the only difference being a constant with  $\gamma^{-2}$  in place of  $\gamma^{-1}$ . This is where the design condition is used a second time. For the levels, Lemma S5.1 gives  $\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c}) = \Pi_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon}) - \Pi_{\mathbf{A}}\mathbf{B}v$ , so that  $\|\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c})\|_n \leq \|\mathbf{b}\|_n + \|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n + \lambda_{\max}(\widehat{\Sigma})^{1/2} \|v\|_2$ , while  $\mathbf{A}^\top \mathbf{A}/n$  is a principal submatrix of  $\widehat{\Sigma}$ , whence  $\|\widehat{\mathbf{c}} - \mathbf{c}\|_2^2 \leq \|\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c})\|_n^2 / (\gamma/2)$ . Finally the components of a functional coefficient depend on distinct coordinates of  $u$  and the basis is orthonormal, so  $\|\widehat{\beta}_\ell - \beta_\ell\|_{L_2([0,1]^q)} \leq |\widehat{c}_\ell - c_\ell| + \sum_m \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}$ , with  $q$  fixed. The same argument applies verbatim under the conditions of Theorem 1.  $\square$

### Proof of Theorem 1

*A nonasymptotic bound.* Apply Lemma S6.1 with  $\bar{\boldsymbol{\beta}} = (\mathbf{c}, \boldsymbol{\theta}^{*,(M)})$ , the best  $M$  term approximation of the oracle. Then  $\bar{M} \leq M$  and  $\bar{\mathbf{b}} = \mathbf{b} + \mathbf{B}(\boldsymbol{\theta}^* -$

$\boldsymbol{\theta}^{*,(M)}$ ), so that, by the Cauchy-Schwarz inequality, Lemma 3 and (S6.2),

$$\|\bar{\mathbf{b}}\|_n^2 \leq 2\|\mathbf{b}\|_n^2 + 2\lambda_{\max}(\widehat{\Sigma}) \left\| \boldsymbol{\theta}^* - \boldsymbol{\theta}^{*,(M)} \right\|_2^2 \leq 2\|\mathbf{b}\|_n^2 + \frac{2\Lambda C_\tau^2 \tau}{2-\tau} M^{1-2/\tau}.$$

Substituting in (S6.1), on the intersection of  $\mathcal{T}$  with  $\{\lambda_{\min}(\widehat{\Sigma}) \geq \gamma/2\}$ ,

$$\left\| \widehat{f} - f \right\|_n^2 \leq 3 \min_{1 \leq M \leq d} h(M) + 120 \|\mathbf{b}\|_n^2 + 3 \|\Pi_{\mathbf{A}} \boldsymbol{\epsilon}\|_n^2, \quad h(M) = \frac{64\lambda^2}{\widetilde{\gamma}} M + \frac{40\Lambda C_\tau^2 \tau}{2-\tau} M^{1-2/\tau}.$$

The function  $h$  is strictly convex on the positive half line, since  $1 - 2/\tau < 0$ , and  $h'$  vanishes at  $M^* = \{5\Lambda C_\tau^2 \widetilde{\gamma} / (8\lambda^2)\}^{\tau/2}$ , where  $h(M^*) = 2(2 - \tau)^{-1}(64\lambda^2/\widetilde{\gamma})M^*$ ; moreover  $h(\lceil M^* \rceil) \leq h(M^*) + 64\lambda^2/\widetilde{\gamma} \leq 2h(M^*)$  whenever  $M^* \geq 1$ , because  $(2 - \tau)/2 < 1$ .

*The resolution.* With  $\lambda_n^2 \asymp \sigma^2 B_X^2 C_U J_n/n$ ,  $\widetilde{\gamma} \in [\gamma/2, \Lambda]$  and the other constants fixed,  $M_n^* \asymp \lambda_n^{-\tau} \asymp (n/J_n)^{\tau/2}$  and  $h(M_n^*) \asymp \lambda_n^2 M_n^* \asymp (J_n/n)^{(2-\tau)/2}$ . Writing  $J_n = \lceil c \log_2 n \rceil$ , so that  $2^{J_n} \asymp n^c$ , the three inequalities of the window (3.2) are, in order:  $M_n^* \leq d_n$ , which asks  $(n/J_n)^{\tau/2} \lesssim pq n^c$ , that is  $c \geq \tau/2$  up to logarithms; that the bias not dominate,  $n^{-2cs'} \lesssim (J_n/n)^{(2-\tau)/2}$ , that is  $c \geq (2 - \tau)/(4s')$ ; and the requirement of Proposition 2(ii),  $n^{c-1} \log n \rightarrow 0$ , that is  $c < 1$ . The term  $\sigma^2 p/n$  is of smaller order because  $(2 - \tau)/2 < 1$ . The window is nonempty if and only if  $\tau/2 < 1$ , which always holds, and  $(2 - \tau)/(4s') < 1$ , that is  $s' > (2 - \tau)/4$ .

*Under Assumption 5 the hypothesis is automatic.* By Lemma 3 the envelope holds with  $\tau = (s+1/2)^{-1}$ , whence  $2 - \tau = 4s/(2s+1)$ ,  $(2 - \tau)/2 =$

$2s/(2s+1)$  and  $(2-\tau)/4 = s/(2s+1)$ , which is the condition  $s' > s/(2s+1)$  of the statement; for  $\pi \geq 2$  it holds for every  $s > 0$ , and for  $\pi < 2$  it reads  $2s^2/(2s+1) > 1/\pi - 1/2$ . The rate for the components is the argument of Corollary 1 with  $\left\| \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^* \right\|_2 \leq \|\bar{v}\|_2 + \left\| \boldsymbol{\theta}^* - \boldsymbol{\theta}^{*,(M)} \right\|_2$  and Lemma S6.1, the only difference being an extra factor  $\tilde{\gamma}^{-1}$  in the first term, which does not change the order. Finally  $2s/(2s+1) \geq 2s'/(2s'+1)$ , with equality only when  $s' = s$ , by monotonicity of  $x \mapsto 2x/(2x+1)$  and  $s' \leq s$ , so the rate is never worse than that of Theorem 3 and is strictly better when  $\pi < 2$ .  $\square$

**Remark S6.1** (What compressibility buys). Read at the oracle, Theorem 2(ii) has  $s_0 = d$  and delivers the rate of [linear approximation](#); read at a sparse comparator, through Lemma S6.1, it delivers the rate of Theorem 1. It is the same inequality in both cases, and neither says anything about  $\text{supp}(\widehat{\boldsymbol{\theta}})$ : no minimal signal and no irrepresentability condition is used, and what is gained is a rate governed by the effective dimension  $M_n^*$  rather than by  $d_n$ .

## S7 Recovery of the block structure by thresholding

Throughout this section  $N_{\ell m}$ ,  $\hat{N}_{\ell m}$ ,  $S$ ,  $\hat{S}(t)$ ,  $\rho_n$  and  $\delta_n$  are as in Section 3.6 of the paper,  $\delta = \min_{(\ell, m) \in S} N_{\ell m}$  for a fixed sample size, and

$$\Delta = \max_{\ell, m} \|\hat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}$$

is the largest error over the blocks. The first statement is deterministic and knows nothing about the lasso.

**Lemma S7.1** (Thresholding under a uniform block error). *Let  $\{g_{\ell m}\}$  and  $\{\hat{g}_{\ell m}\}$  be any two families of functions in  $L_2[0, 1]$  indexed by the blocks, with  $N_{\ell m}$ ,  $\hat{N}_{\ell m}$ ,  $S$ ,  $\hat{S}(t)$ ,  $\Delta$  and  $\delta$  defined from them as above, and let  $D \geq \Delta$ . Then, for every  $t \geq 0$ ,*

$$\{(\ell, m) : N_{\ell m} > t + D\} \subseteq \hat{S}(t) \subseteq \{(\ell, m) : N_{\ell m} > t - D\}. \quad (\text{S7.1})$$

*In particular: (i) if  $t \geq D$ , then  $\hat{S}(t) \subseteq S$ , with no use of  $\delta$ ; (ii) if  $t < \delta - D$ , then  $S \subseteq \hat{S}(t)$ ; and (iii) if  $D \leq t < \delta - D$ , then  $\hat{S}(t) = S$ , and such a  $t$  exists if and only if  $\delta > 2D$ .*

*Proof.* By the triangle inequality in  $L_2[0, 1]$ , for every block,  $|\hat{N}_{\ell m} - N_{\ell m}| \leq \|\hat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]} \leq \Delta \leq D$ . If  $N_{\ell m} > t + D$ , then  $\hat{N}_{\ell m} \geq N_{\ell m} - D > t$ , which is the left inclusion of (S7.1); if  $\hat{N}_{\ell m} > t$ , then  $N_{\ell m} \geq \hat{N}_{\ell m} - D > t - D$ , which is the right one. For (i),  $t - D \geq 0$  and the right inclusion give

$\widehat{S}(t) \subseteq \{N_{\ell m} > 0\} = S$ . For (ii), every block of  $S$  has  $N_{\ell m} \geq \delta > t + D$ , and the left inclusion applies. Part (iii) is the conjunction of the two, and  $[D, \delta - D)$  is nonempty exactly when  $D < \delta - D$ .  $\square$

**Remark S7.1** (The window is exact). The condition  $\delta > 2D$  is not an artifact of the proof. Take two blocks, one active with  $N_{11} = 2D$  and one null with  $N_{21} = 0$ , and fitted norms  $\widehat{N}_{11} = \widehat{N}_{21} = D$ , which is compatible with  $\Delta \leq D$ : the two statistics tie and no threshold separates them. This says that the argument does not extend below  $\delta = 2D$ , not that no procedure does, which would be a lower bound and is not proved here.

### Proof of Corollary 2

A *nonasymptotic bound on  $\Delta$* . Work at  $J = J_n$  and  $\lambda = \lambda_n$ , and intersect the four events of the proof of Corollary S5.1:  $\mathcal{T}$ , of probability at least  $1 - \alpha$ , by Lemma S5.2;  $\{\widehat{\sigma}_{\max}^2 \leq 2B_X^2 C_U\}$ , by Lemma S5.3, which makes  $\lambda_n$  admissible;  $\{\lambda_{\min}(\widehat{\Sigma}) \geq \gamma/2\}$ , by Proposition 2(ii), which with (S5.2) gives  $\widetilde{\gamma} \geq \gamma/2$ ; and  $\{\|\mathbf{b}\|_n^2 \leq \mathcal{B}_{J_n}^2/\alpha'\}$ , by Markov's inequality and Lemma 2. On this event Theorem 2(ii) with  $s_0 \leq d_n$  gives

$$\|v\|_2^2 \leq \frac{64\lambda_n^2 d_n}{\widetilde{\gamma}^2} + \frac{16\|\mathbf{b}\|_n^2}{\widetilde{\gamma}} \leq \frac{256\lambda_n^2 d_n}{\gamma^2} + \frac{32\mathcal{B}_{J_n}^2}{\alpha'\gamma}.$$

For a fixed block, the orthonormality of the wavelets of level below  $J_n$  gives  $\|\widehat{g}_{\ell m} - \Pi_J g_{\ell m}\|_{L_2[0,1]} = \|v_{\ell m,\cdot}\|_2 \leq \|v\|_2$ , and the first display of Lemma

2 bounds  $\|\Pi_J g_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2$  by  $C_g^2(1 - 2^{-2s'})^{-1}2^{-2J_n s'}$ . By  $(x + y)^2 \leq 2x^2 + 2y^2$ ,

$$\Delta^2 \leq D_n^2 = \frac{512 \lambda_n^2 d_n}{\gamma^2} + \frac{64 \mathcal{B}_{J_n}^2}{\alpha' \gamma} + \frac{2C_g^2}{1 - 2^{-2s'}} 2^{-2J_n s'} \quad (\text{S7.2})$$

with probability at least  $1 - \alpha - \alpha' - \mathbb{P}\{\lambda_{\min}(\widehat{\Sigma}) < \gamma/2\} - \mathbb{P}\{\widehat{\sigma}_{\max}^2 > 2B_X^2 C_U\}$ , the last two terms tending to zero by Proposition 2(ii) and Lemma S5.3. By Lemma S7.1(iii), on that event  $\widehat{S}(t) = S$  for every  $t$  with  $D_n \leq t < \delta - D_n$ , a window that is nonempty whenever  $\delta > 2D_n$ . The bound  $D_n$  depends on the components only through the constants of Assumption 5, and the two residual probabilities do not depend on them at all, so everything above holds uniformly over the class, which is what allows the target to vary with  $n$  in Assumption 6.

*The order of  $D_n$ .* At the resolution of Theorem 3 the three terms of (S7.2) are at most of order  $\rho_n^2$ : the first because  $\lambda_n^2 d_n \asymp pq2^{J_n} J_n/n$ , which is the quantity shown to be of order  $(\log n/n)^{2s'/(2s'+1)}$  in the proof of Theorem 3, and the other two because  $\mathcal{B}_{J_n}^2 \asymp 2^{-2J_n s'} \leq (\log n/n)^{2s'/(2s'+1)}$  by the definition of  $J_n$ . Hence  $D_n \leq c(\alpha, \alpha')\rho_n$  with  $c(\alpha, \alpha') < \infty$ . Given  $\epsilon > 0$ , take  $\alpha = \alpha' = \epsilon/4$  and  $n$  large enough that the two residual probabilities add up to less than  $\epsilon/2$ ; then  $\mathbb{P}\{\Delta > c(\epsilon/4, \epsilon/4)\rho_n\} < \epsilon$ , that is  $\Delta = O_p(\rho_n)$ .

*Part (i).* If  $t_n/\rho_n \rightarrow \infty$ , then  $\mathbb{P}(\Delta \leq t_n) \rightarrow 1$ , and on that event Lemma S7.1(i) with  $D = \Delta$  gives  $\widehat{S}(t_n) \subseteq S$ . No property of  $\delta_n$  has been used.

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*Part (ii).* If  $S$  is empty, part (ii) is part (i). Otherwise, the condition  $\limsup_n t_n/\delta_n < 1$  gives  $\eta > 0$  and  $n_0$  with  $\delta_n - t_n \geq \eta \delta_n$  for  $n \geq n_0$ , and Assumption 6 gives  $(\delta_n - t_n)/\rho_n \geq \eta \delta_n/\rho_n \rightarrow \infty$ . Hence  $P(\Delta \leq t_n) \rightarrow 1$  and  $P(\Delta < \delta_n - t_n) \rightarrow 1$ , and on the intersection of the two events Lemma S7.1(iii) with  $D = \Delta$  gives  $\widehat{S}(t_n) = S$ . For the existence,  $t_n = (\rho_n \delta_n)^{1/2}$  has  $t_n/\rho_n = (\delta_n/\rho_n)^{1/2} \rightarrow \infty$  and  $t_n/\delta_n = (\rho_n/\delta_n)^{1/2} \rightarrow 0$ .  $\square$

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